Part A — Theory

1. **difference between WHERE and HAVING. Give a business example for each.**

WHERE vs HAVING

WHERE filters individual rows before any aggregation happens. It works on raw data and removes unwanted records before calculations like SUM or COUNT are applied.

Business example: A UK e-commerce company wants to calculate total revenue from completed orders only. The analyst uses WHERE order_status = 'completed' to exclude cancelled orders before summing revenue.

HAVING, however, filters results after aggregation. It is used with GROUP BY when you want to filter grouped data.

Business example: Management wants to see only regions generating more than £100,000 in revenue. After grouping revenue by region, the analyst uses HAVING SUM(revenue) > 100000 to filter high-performing regions.

2. **Explain INNER JOIN vs LEFT JOIN in plain English. When would a business analyst choose LEFT JOIN?**

INNER JOIN vs LEFT JOIN

An **INNER JOIN** returns only records that exist in both tables. If there is no match, the record is excluded.

Business example: If a company wants to analyse purchasing behaviour, an INNER JOIN between customers and orders will return only customers who have placed at least one order.

A **LEFT JOIN** returns all records from the left table, even if there is no match in the right table. Missing matches appear as NULL values.

Business example: If the marketing team wants to calculate conversion rate (registered users vs. buyers), a LEFT JOIN keeps all registered customers, including those who never purchased.

3. **What does "data grain" mean? In this dataset, what does 1 row represent and why does it matter for SUM/COUNT KPIs**?

What is Data Grain?

Data grain defines what one row represents in a dataset. For example, in a retail dataset, one row might represent one product line within an order rather than one full order.

Business example: If each row represents one product per order, then counting order_id directly will count product lines, not unique orders. To calculate total orders correctly, the analyst must use COUNT(DISTINCT order_id).

Understanding grain is essential because KPIs like revenue, order count, and AOV depend on it. Misunderstanding grain leads to inflated or incorrect business metrics.

4. **Why can joins create duplicated rows and inflate revenue? Describe one safe technique to prevent this (hint: pre-aggregate the 'many' side).**

Why Joins Can Inflate Revenue

Joins can duplicate rows when there is a one-to-many relationship. For example, one order may have multiple products, and joining orders with order_items will create multiple rows for the same order.

Business example: If Order 101 has 3 items and revenue from the orders table is £300, after the join it may appear three times. If summed incorrectly, revenue becomes £900 instead of £300.

To prevent this, analysts should pre-aggregate the many-side table (e.g., sum item revenue by order_id first) before joining. This ensures accurate revenue reporting.

5. **Explain how DISTINCT helps with data cleanliness and when it can be misused.**

DISTINCT – Use and Misuse

DISTINCT removes duplicate rows and is commonly used to count unique values.

Business example: To calculate the number of unique customers who made a purchase, the analyst uses COUNT(DISTINCT customer_id) to avoid double-counting customers with multiple orders.

However, DISTINCT can be misused to hide duplication problems caused by incorrect joins.

Business example: If revenue is inflated due to a bad join, using DISTINCT may reduce duplicate rows but will not fix incorrect SUM calculations. The correct solution is to fix the join logic, not rely on DISTINCT.

1. Introduction

This report analyses the UK Retail transactional dataset for 2024 using SQL queries to identify key business insights related to revenue performance, customer behaviour, marketing channels, refund trends, and operational efficiency.

The dataset contains 1,235 order item records representing 542 completed orders from 202 unique customers. Each record includes details such as order date, product category, quantity, price, customer information, and refund data.

The total revenue generated from completed orders in the dataset is approximately £95,953.70, with total recorded refunds of £3,235.11. The objective of this analysis is to transform transactional data into meaningful insights that can support data-driven decision-making.

2. Data Preparation and Cleaning

Before performing the analysis, several data quality checks were conducted. Duplicate records, missing values, and inconsistent text fields were examined to ensure accurate results. Fields such as customer names, cities, and product categories were standardised using SQL cleaning techniques including trimming spaces and converting text to lowercase.

Revenue calculations were also validated by confirming that line_amount = unit_price × quantity for each order item. Additionally, the dataset was normalised into structured tables such as customers, products, orders, order items, and refunds, which improves query performance and data integrity.

3. Key Business Insights

Revenue Performance

The dataset generated a total completed revenue of £95,953.70 across 542 completed orders, indicating an average order value of approximately £177.

Revenue is heavily concentrated in a few product categories. The Electronics category generated the highest revenue at £30,000, followed by Sports (£21,415) and Home products (£18,552).

Category	Revenue
Electronics	£30,000
Sports	£21,415
Home	£18,552
Beauty	£15,260
Grocery	£6,320

These results indicate that Electronics and Sports products are the strongest revenue drivers and should remain a focus for marketing and inventory planning.

Customer Behaviour

The analysis identified 202 unique purchasing customers, generating 542 completed orders. This suggests that many customers made multiple purchases, indicating a healthy level of repeat buying behaviour.

Customer tier analysis shows that higher-tier customers tend to contribute more revenue per order. This suggests that targeted loyalty programs and personalised promotions for high-value customers could significantly increase long-term revenue.

Acquisition Channel Performance

Marketing channel analysis shows that Organic traffic generated the highest revenue, contributing approximately £30,567 in sales. Paid search and paid social campaigns also contributed significantly.

Acquisition Channel	Revenue
Organic	£30,567
Paid Search	£21,776
Paid Social	£17,497
Email	£12,891
Affiliate	£7,666

Although paid channels drive traffic, organic acquisition generates the highest value. This suggests that improving SEO and organic marketing strategies could provide strong long-term growth.

Refund Trends

The dataset shows a total refund value of £3,235, representing a small percentage of total revenue. However, certain product categories have higher refund levels than others.

Higher refund rates may indicate issues related to product quality, inaccurate product descriptions, or customer expectations. Monitoring refund patterns can help identify problematic products or suppliers.

Reducing refund rates by improving product descriptions and quality checks could directly increase net revenue.

Delivery Performance

Shipping performance analysis shows that 45 completed orders experienced delivery delays greater than five days. Extended delivery times can negatively affect customer satisfaction and increase the likelihood of returns.

Improving logistics operations, selecting reliable delivery partners, and monitoring shipping performance metrics can help improve customer experience and reduce operational risks.

4. Business Recommendations

Based on the insights from this analysis, several strategic recommendations can be made:

1. Focus on high-revenue product categories
Increase marketing and product availability for Electronics and Sports categories, which generate the highest sales.

2. Strengthen customer retention strategies
Introduce loyalty programs or personalised promotions for repeat and high-tier customers.

3. Invest in organic marketing channels
Since organic traffic generates the highest revenue, improving SEO and content marketing strategies could increase profitability.

4. Monitor refund patterns
Investigate categories with higher refund amounts and improve product descriptions or supplier quality.

5. Improve logistics efficiency
Reduce delivery delays by optimising logistics partners and monitoring shipping performance metrics.

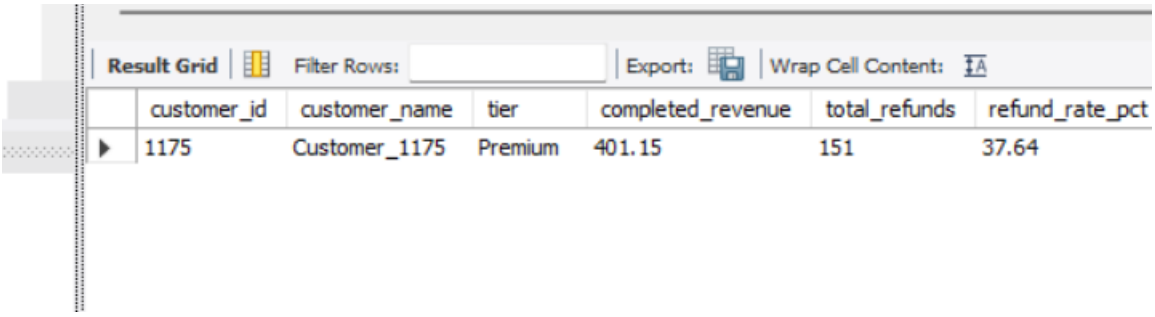
5. Conclusion

The SQL-based analysis of the UK Retail dataset demonstrates how transactional data can be transformed into meaningful business insights. With total revenue exceeding £95,000 from over 500 completed orders, the business shows strong sales potential.

Key opportunities for improvement include optimising marketing channels, focusing on high-performing product categories, reducing refund rates, and improving delivery performance.

Data-driven insights such as these enable businesses to make more informed strategic decisions and improve overall operational efficiency.

Screen shot of analysis



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_name	tier	completed_revenue	total_refunds	refund_rate_pct
1175	Customer_1175	Premium	401.15	151	37.64

Result Grid | Filter Rows: | Export:

SIGNUP_MONTH	CUSTOMERS	REVENUE_FIRST_90DAYS
2023-02	9	0
2023-03	13	0
2023-04	9	0
2023-05	14	0
2023-06	5	0
2023-07	9	0
2023-08	6	0
2023-09	9	0
2023-10	11	0
2023-11	9	9.75
2023-12	11	1129.02
2024-01	6	377.64
2024-02	7	282.43
2024-03	8	717.74
2024-04	13	1126.16
2024-05	15	2938.96
2024-06	5	567.17
2024-07	5	552.87
2024-08	5	234.87
2024-09	8	459.15
2024-10	9	1933.65
2024-11	9	131.5
2024-12	8	784.: 131.5

Result Grid | Filter

normalized_revenue
95953.7

Result Grid

raw_revenue
95953.7

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

order_id	customer_id	customer_name	city	product_name	unit_price	quantity	line_amount	total_amount
900203	1192	Customer_1192	Newport	Electronics_Item_2021	540.66	2	1081.32	1081.32
900634	1122	Customer_1122	London	Electronics_Item_2021	540.66	2	1081.32	1081.32
900597	1010	Customer_1010	Preston	Electronics_Item_2021	459.56	2	919.12	919.12
900016	1021	Customer_1021	Swansea	Electronics_Item_2026	214.17	3	642.51	642.51
900622	1089	Customer_1089	Swansea	Sports_Item_2032	188.74	3	566.22	566.22
900003	1121	Customer_1121	Reading	Electronics_Item_2021	540.66	1	540.66	540.66
900143	1203	Customer_1203	Leicester	Electronics_Item_2021	540.66	1	540.66	540.66
900146	1066	Customer_1066	London	Electronics_Item_2021	540.66	1	540.66	540.66
900360	1162	Customer_1162	Liverpool	Electronics_Item_2021	540.66	1	540.66	540.66
900538	1217	Customer_1217	London	Electronics_Item_2021	540.66	1	540.66	540.66
900099	1131	Customer_1131	Aberdeen	Electronics_Item_2021	540.66	1	540.66	540.66
900406	1143	Customer_1143	Exeter	Electronics_Item_2021	540.66	1	540.66	540.66
900636	1172	Customer_1172	Coventry	Electronics_Item_2021	540.66	1	540.66	540.66
900571	1150	Customer_1150	Reading	Electronics_Item_2021	540.66	1	540.66	540.66
900613	1063	Customer_1063	London	Electronics_Item_2021	540.66	1	540.66	540.66
900547	1114	Customer_1114	Cardiff	Electronics_Item_2021	540.66	1	540.66	540.66
900550	1140	Customer_1140	Swansea	Electronics_Item_2021	513.63	1	513.63	513.63
900095	1218	Customer_1218	Edinburgh	Electronics_Item_2021	513.63	1	513.63	513.63
900290	1051	Customer_1051	Newcastle	Electronics_Item_2021	513.63	1	513.63	513.63
900423	1104	Customer_1104	Cardiff	Electronics_Item_2021	513.63	1	513.63	513.63

Schemas

elected

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	product_name	line_amount	calculated_amount	difference

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_rows	customer_id_not_null	customer_name_not_null	city_not_null	product_not_null
1157	1157	1157	1157	1157

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch row

order_id	customer_name	region	tier	order_date	shipping_days	order_value
900340	Customer_1182	London	Premium	2024-08-13	8	516.17
900051	Customer_1029	North East	Standard	2024-06-26	7	166.69
900105	Customer_1025	London	Standard	2024-07-13	7	78.38
900377	Customer_1137	North West	Premium	2024-06-23	7	171.04
900603	Customer_1074	London	Standard	2024-02-12	7	92.7
900001	Customer_1115	North East	Standard	2024-07-21	6	41.27
900021	Customer_1020	North West	Standard	2024-04-24	6	23.97
900024	Customer_1149	Scotland	Premium	2024-07-30	6	147.38
900042	Customer_1087	North East	Standard	2024-04-27	6	139.78
900054	Customer_1219	London	Standard	2024-11-21	6	180.73
900069	Customer_1010	North West	Standard	2024-06-26	6	69.89
900119	Customer_1177	North East	Standard	2024-04-10	6	70.57
900132	Customer_1138	South East	Premium	2024-11-18	6	52.04
900180	Customer_1055	London	Standard	2024-05-16	6	67.27
900216	Customer_1214	Scotland	Standard	2024-07-27	6	297.11
900242	Customer_1023	Scotland	Standard	2024-01-21	6	29.25
900320	Customer_1101	London	Standard	2024-06-29	6	85.02
900322	Customer_1143	South West	Standard	2024-02-21	6	74.05
900365	Customer_1203	Midlands	Premium	2024-09-18	6	63.5
900374	Customer_1034	Scotland	Standard	2024-04-04	6	117.76

Result Grid | Filter Rows: | Export:

category	revenue	refunded	refund_rate
Electronics	34230.34	799.13	2.33%
Sports	24591.64	602.38	2.45%
Home	21634.76	752.28	3.48%
Beauty	16771.11	643.06	3.83%
Grocery	7252.81	255.06	3.52%

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

acquisition_channel	customers_acquired	completed_revenue	refund_amount	net_revenue
Organic	123	30567.91	776.09	29791.82
Paid Search	93	21776.36	989.26	20787.1
Paid Social	78	17497.37	456.37	17041
Email	62	12891.45	259.66	12631.79
Affiliate	36	7666.64	523.97	7142.67
Referral	40	5553.97	229.76	5324.21

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

tier	Customers	completed_orders	revenue	AOV	pct_customers_purchased
Premium	67	158	25245.63	159.78	98.51%
Standard	141	384	70708.07	184.14	96.45%

Result Grid | Filter Rows: | Export: | Wra

region	completed_orders	total_revenue	AOV
London	146	25107.78	171.97
Wales	68	15227.68	223.94
South East	65	11071.35	170.33
Midlands	58	10633.59	183.34
Scotland	65	10445.01	160.69
North West	51	9145.14	179.32
South West	47	7595.85	161.61
North East	42	6727.3	160.17

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch

CUSTOMER_NAME	REGION	TIER	order_status	total_revenue
Customer_1122	London	Standard	Completed	2029.9
Customer_1131	Scotland	Standard	Completed	1680.05
Customer_1182	London	Premium	Completed	1648.06
Customer_1114	Wales	Standard	Completed	1642.05
Customer_1007	London	Standard	Completed	1398.31
Customer_1010	North West	Standard	Completed	1352.85
Customer_1217	London	Standard	Completed	1329.27
Customer_1026	London	Standard	Completed	1291.78
Customer_1089	Wales	Standard	Completed	1274.27
Customer_1016	South East	Standard	Completed	1250.89

Result Grid | Filter Rows:

AOV_2024
177.03634686346876

Result Grid | Filter Rows:

total_completed_revenue
95953.70000000007

Result Grid | Filter Rows:

unique_customers_2024
208